

Using the KSP tools, calculate delta-v for a rocket with Isp 300s, wet mass 10 tonnes, dry mass 2 tonnes.

Claude responded: Delta-v: ~4,735 m/s

Loaded tools, used ksp-mission-control integration

Loaded tools, used ksp-mission-control integration

Delta-v: ~4,735 m/s

Using the Tsiolkovsky rocket equation: $\Delta v = I_{sp} \times g_0 \times \ln(m_{wet} / m_{dry}) = 300 \times 9.80665 \times \ln(5) \approx 4,735 \text{ m/s}$.

You said: What is the orbital velocity for a circular orbit 100 km above Kerbin?

What is the orbital velocity for a circular orbit 100 km above Kerbin? Kerbin's gravitational parameter is $3.5316 \times 10^{12} \text{ m}^3/\text{s}^2$ and its radius is 600,000 m.

Claude responded: The tool needs body mass, not μ .

Loaded tools

Loaded tools

The tool needs body mass, not μ . Converting: $M = \mu/G = 3.5316 \times 10^{12} / 6.674 \times 10^{-11} \approx 5.2916 \times 10^{22} \text{ kg}$.

calculate_orbital_velocity

calculate_orbital_velocity

Orbital velocity: ~2,246 m/s

For a 100 km circular orbit around Kerbin (orbital radius = 700 km), $v = \sqrt{\mu/r} = \sqrt{(3.5316 \times 10^{12} / 700,000)} \approx 2,246 \text{ m/s}$.

You said: What is the orbital period of a circular orbit 100 km above Kerbin (same parameters as above)?

What is the orbital period of a circular orbit 100 km above Kerbin (same parameters as above)?

Claude responded: Orbital period: ~1,958 s ($\approx 32 \text{ min } 38 \text{ s}$)

Loaded tools, used ksp-mission-control integration

Loaded tools, used ksp-mission-control integration

Orbital period: ~1,958 s ($\approx 32 \text{ min } 38 \text{ s}$)

$T = 2\pi\sqrt{r^3/\mu} = 2\pi \times \sqrt{(700,000^3 / 3.5316 \times 10^{12})} \approx 1,958 \text{ seconds}$.

You said: Calculate the Hohmann transfer from a 100 km circular orbit to a 250 km orbit around Kerbin.

Calculate the Hohmann transfer from a 100 km circular orbit to a 250 km orbit around Kerbin.

Claude responded: Hohmann transfer 100 km → 250 km around Kerbin:

Loaded tools, used ksp-mission-control integration

Loaded tools, used ksp-mission-control integration

Hohmann transfer 100 km → 250 km around Kerbin:

- Burn 1 (prograde at 100 km): 106.2 m/s
- Burn 2 (circularize at 250 km): 101.1 m/s
- Total Δv : 207.3 m/s
- Transfer time: ~1,141 s ($\approx$ 19 min) — half the elliptical period

You said: What is the escape velocity from Kerbin's surface (altitude 0)?

What is the escape velocity from Kerbin's surface (altitude 0)?

Claude responded: Escape velocity from Kerbin's surface: ~3,431 m/s

Loaded tools, used ksp-mission-control integration

Loaded tools, used ksp-mission-control integration

Escape velocity from Kerbin's surface: ~3,431 m/s

$v_{\text{esc}} = \sqrt{2\mu/r} = \sqrt{2 \times 3.5316 \times 10^{12} / 600,000} \approx 3,431 \text{ m/s}$.

You said: What altitude is Kerbin's synchronous orbit?

What altitude is Kerbin's synchronous orbit? Kerbin's sidereal rotation period is 21,549 seconds.

Claude responded: Kerbin's synchronous (keostationary) orbit:

Loaded tools, used ksp-mission-control integration

Loaded tools, used ksp-mission-control integration

Kerbin's synchronous (keostationary) orbit:

- Altitude: ~2,863 km above Kerbin's surface
- Orbital velocity: ~1,010 m/s
- Orbital radius: ~3,463 km from Kerbin's center

Matches the well-known KSP keostationary altitude of ~2,863.33 km.

You said: What is the CommNet range between two RA-15 relay antennas, each with a power rating of 1.

What is the CommNet range between two RA-15 relay antennas, each with a power rating of 1.5×10^8 W?

Claude responded: CommNet range between two RA-15 relays: 150,000,000 m (150,000 km)

Loaded tools, used ksp-mission-control integration

Loaded tools, used ksp-mission-control integration

CommNet range between two RA-15 relays: 150,000,000 m (150,000 km)

Range = $\sqrt{(P_1 \times P_2)} = \sqrt{(1.5 \times 10^8 \times 1.5 \times 10^8)} = 1.5 \times 10^8$ m. With identical antennas the range simply equals the antenna power rating.

You said: CommNet range between an HG-5 (power 5×10^6 W) and an RA-2 (power 2×10^9 W).

CommNet range between an HG-5 (power 5×10^6 W) and an RA-2 (power 2×10^9 W).

Claude responded: CommNet range HG-5 ↔ RA-2: 100,000,000 m (100,000 km)

calculate_commnet_range

calculate_commnet_range

CommNet range HG-5 ↔ RA-2: 100,000,000 m (100,000 km)

Range = $\sqrt{(5 \times 10^6 \times 2 \times 10^9)} = \sqrt{(10^{16})} = 10^8$ m. The weak HG-5 bottlenecks the link — pairing it with a much stronger antenna only modestly extends range over the geometric mean.

You said: Using the KSP tools, what are my current funds, science points, and reputation?

Using the KSP tools, what are my current funds, science points, and reputation?

Claude responded: Current career state:

Loaded tools, used ksp-mission-control integration

Loaded tools, used ksp-mission-control integration

Current career state:

- Funds: $\sqrt{504,662}$
- Science: 20.5
- Reputation: 213.9

You said: Show me my tech tree — which nodes are unlocked and what's available to research next?

Show me my tech tree — which nodes are unlocked and what's available to research next?

Claude responded: You have 13 nodes unlocked and 20.

Loaded tools, used ksp-mission-control integration

Loaded tools, used ksp-mission-control integration

You have 13 nodes unlocked and 20.5 science to spend — not enough to research any new node yet (cheapest locked node is 45 science).

Unlocked (13)

By tier:

- Tier 1 (5): Basic Rocketry, Engineering 101
- Tier 2 (15-20): Survivability, Stability, General Rocketry
- Tier 3 (45): Basic Science, Flight Control, Advanced Rocketry, General Construction
- Tier 4 (90): Advanced Flight Control, Heavy Rocketry, Fuel Systems

Plus Start (0).

Locked — cheapest tiers to target next

- 45 sci: Aviation (only locked node at this tier — first jet engine, wings, cockpit)
- 90 sci: Propulsion Systems, Space Exploration, Landing (LES, landing legs, large chutes), Aerodynamic Systems, Electrics (probe core, solar panels)
- 160 sci: Command Modules, Heavier Rocketry, Precision Engineering, Advanced Landing, Adv Electrics, Specialized Control, etc.
- 300+: Unmanned Tech, Nuclear Propulsion, Scanning Tech, Electronics, Ion Propulsion (550), RTGs (1000), etc.

Suggested next research priorities (depending on play style):

- Electrics (90) — probe core + solar panels unlocks unmanned probes and long-duration missions
- Landing (90) — Launch Escape System, larger landing legs, big chutes for safer Mun/Minmus landings
- Space Exploration (90) — ladders, rover wheels, deployed science for surface ops

You need to earn at least 24.5 more science to afford the first 45-cost node, or 69.5 more for a 90-cost node.

You said: List all rocket engines available in my current game.

List all rocket engines available in my current game.

Claude responded: 12 engines available (all purchased).

Loaded tools, used ksp-mission-control integration

Loaded tools, used ksp-mission-control integration

12 engines available (all purchased).

Liquid Fuel Engines (LFO)

Engine	Thrust (vac)	Isp (vac/asl)	Mass	Notes
LV-909 "Terrier"	60 kN	345/85 s	0.5 t	Vacuum-optimized
LV-T30 "Reliant"	240 kN	310/265 s	1.25 t	No gimbal
LV-T45 "Swivel"	215 kN	320/250 s	1.5 t	Gimbaled
Mk-55 "Thud"	120 kN	305/275 s	0.9 t	Radial mount
RE-L10 "Poodle"	250 kN	350/90 s	1.75 t	Best vac Isp
RE-I5 "Skipper"	650 kN	320/280 s	3.0 t	Heavy lifter
LV-TX87 "Bobcat"	400 kN	310/290 s	2.0 t	Sea-level optimized
Solid Rocket Boosters (SRB)				
Booster	Thrust (vac)	Isp (vac/asl)	Wet/Dry mass	Fuel
RT-5 "Flea"	192 kN	165/140 s	1.5 / 0.45 t	140
RT-10 "Hammer"	227 kN	195/170 s	3.56 / 0.75 t	375
BACC "Thumper"	300 kN	210/175 s	7.65 / 1.5 t	820
S1 "Kickback"	670 kN	220/195 s	24 / 4.5 t	2600
THK "Pollux"	1300 kN	225/200 s	51.5 / 8.0 t	5800

Notable gaps (locked tech): Mainsail, Mammoth (very heavy rocketry), Nerv nuclear (300 sci), Dawn ion (550 sci), Rapier (1000 sci), Spark/Ant/Twitch (precision propulsion, 160 sci).

You said: Get the stats for the LV-T30 Reliant engine.

Get the stats for the LV-T30 Reliant engine.

Claude responded: LV-T30 "Reliant" Liquid Fuel Engine

Loaded tools, used ksp-mission-control integration

Loaded tools, used ksp-mission-control integration

LV-T30 "Reliant" Liquid Fuel Engine

- Tech required: General Rocketry (unlocked, purchased)
- Cost: √1,100
- Mass: 1.25 t (no internal fuel)
- Thrust: 240 kN (vacuum) / 205.2 kN (sea level)
- Isp: 310 s (vacuum) / 265 s (sea level)
- Fuel flow (vac): 0.0789 units/s
- Propellants: LiquidFuel + Oxidizer
- No gimbal (use with a Swivel or fins for control)

TWR vs. Swivel: Reliant has higher thrust (240 vs 215 kN) but lacks gimbal — common pattern is to cluster Reliants around one central Swivel for steering.

You said: List all fuel tanks available

List all fuel tanks available

Claude responded: 18 fuel tank parts available.

get_parts_by_category

get_parts_by_category

18 fuel tank parts available.

LFO tanks — Size 0 (1.25 m / FL-T series)

Tank	Wet/Dry mass	Fuel (LF+Ox)	Tech
FL-T100	0.56 / 0.0625 t	45 + 55	basicRocketry
FL-T200	1.13 / 0.125 t	90 + 110	generalRocketry
FL-T400	2.25 / 0.25 t	180 + 220	advRocketry
FL-T800	4.50 / 0.5 t	360 + 440	fuelSystems

LFO tanks — Size 1.5 (1.875 m / FL-TX series)

Tank	Wet/Dry mass	Fuel (LF+Ox)	Tech
FL-TX220	1.24 / 0.138 t	99 + 121	advRocketry

FL-TX440	2.48 / 0.275 t	198 + 242	fuelSystems
FL-TX900	5.06 / 0.5625 t	405 + 495	fuelSystems
FL-C1000	6.78 / 0.75 t	540 + 660 + 4 SF	fuelSystems — has integrated 32 kN SRB

LFO tanks — Size 2 (2.5 m / Rockomax X200)

Tank	Wet/Dry mass	Fuel (LF+Ox)	Tech
X200-8	4.50 / 0.5 t	360 + 440	fuelSystems
X200-16	9.00 / 1.0 t	720 + 880	fuelSystems
X200-32	18.0 / 2.0 t	1440 + 1760	fuelSystems

All LFO tanks have the standard 9:1 wet:dry mass ratio.

Adapter tanks (also hold fuel)

Adapter	Wet/Dry mass	Fuel	Connects
FL-A150	0.90 / 0.10 t	72 + 88	0.625m ↔ 1.875m
FL-A151S	0.90 / 0.10 t	72 + 88	1.25m ↔ 1.875m (short)
FL-A151L	3.38 / 0.375 t	270 + 330	1.25m ↔ 1.875m (long)
FL-A215	6.75 / 0.75 t	540 + 660	1.875m ↔ 2.5m

Other

- Stratus-V Monoprop Tank (radial): 0.1 / 0.02 t, 20 MP
- FTX-2 External Fuel Duct: transfer line, 0.05 t
- FTE-1 Drain Valve: dumps fuel, 0.01 t

Notable gaps: no Jumbo-64 / Kerbodyne S3/S4 (need very heavy rocketry, high-perf fuel systems), no Oscar-B mini tanks (propulsion systems), no xenon tanks (ion propulsion).

You said: List all command pods available.

List all command pods available.

Claude responded: 7 command pods available (all purchased).

get_parts_by_category

get_parts_by_category

7 command pods available (all purchased).

Crewed pods

Pod	Crew	Dry mass	EC / MP / Ablator	Tech	Cost
Mk1 Command Pod	1	0.71 t	50 / 10 / —	start	√600
Mk1 Lander Can	1	0.51 t	50 / 15 / —	advFlightControl	√1,500
KV-1 "Onion"	1	0.66 t	50 / — / 20	start	√600
KV-2 "Pea"	2	1.31 t	50 / — / 20	flightControl	√2,000
KV-3 "Pomegranate"	3	1.97 t	50 / — / 20	advFlightControl	√3,000
Mk2 Command Pod	2	1.37 t	200 / — / —	advFlightControl	√2,800

Probe cores

Core	Dry mass	EC	Tech	Cost
Stayputnik	0.05 t	10	basicScience	√300

Notes:

- No pod currently has built-in SAS (all sasLevel: 0) — you'll need separate SAS modules or a pilot Kerbal
- All have 5 km internal antenna (not combinable, low data rate)
- KV-series pods come with built-in ablator (no separate heat shield needed for survivable reentry)
- Mk2 Pod is lightest 2-crew option but no monoprop or ablator — pair with shield + RCS
- Stayputnik is your only unmanned option until Electrics (90 sci) unlocks the OKTO probe core

You said: What science experiments have I completed and what's still available?

What science experiments have I completed and what's still available?

Claude responded: Science overview

Used ksp-mission-control integration, loaded tools

Used ksp-mission-control integration, loaded tools

Science overview

- Kerbin: 9/15 subjects fully maxed, ~8.8 sci remaining
- Mun: 6/11 subjects fully maxed, ~75.9 sci remaining
- Total available: ~84.7 sci

Kerbin — what's left (~8.8 sci)

Mostly tapped out. Significant remainders:

- Mystery Goo @ InSpaceHigh — 1.0 sci left
- Mystery Goo @ InSpaceLow — 0.7 sci left
- Materials Lab @ InSpaceLow — 7.0 sci left
- Trivial bits: Goo FlyingLow (0.03), recovery flew/suborbit/orbit (<0.05 each combined)

Mun — what's left (~75.9 sci) — the big payoff

Experiment	Situation	Remaining
Mobile Materials Lab	InSpaceHigh	39.5 sci
Mobile Materials Lab	InSpaceLow	21.0 sci
Mystery Goo	InSpaceLow	9.0 sci
Mystery Goo	InSpaceHigh	6.0 sci
Recovery (FlewBy)	—	0.4 sci

Fully completed (remaining = 0)

Kerbin: Temp & Crew Report @ FlyingLow Shores, Temp/Barometer/Crew Report @ InSpaceLow, Crew Report @ InSpaceHigh, EVA Report @ InSpaceLow (Water & Highlands).

Mun: Barometer/Temp/Crew Report @ InSpaceLow, Crew Report & EVA Report @ InSpaceHigh, EVA Report @ InSpaceLow Midlands.

Suggestions

- Biggest single haul: send a vessel with a Mobile Processing Lab + Goo to a high Mun orbit — that alone is ~45 sci, enough to unlock a 45-cost node with margin.
- Major untapped biomes on the Mun: you've only sampled InSpaceLow/Midlands for EVA — landing in different biomes (Highlands, Craters, Poles, etc.) opens many fresh subjects not listed here because they haven't been attempted.
- You have no science from Minmus or any other body — Minmus is easier than the Mun (lower gravity, flat biomes) and would unlock huge science.

You said: What vessels do I currently have in flight or on the ground?

What vessels do I currently have in flight or on the ground?

Claude responded: 1 active vessel:

Loaded tools, used ksp-mission-control integration

Loaded tools, used ksp-mission-control integration

1 active vessel:

- Nedory's Debris (Ship, with Nedory Kerman aboard) — orbiting Kerbin, 89 km × 81 km at near-zero inclination

A few flags worth raising:

- Despite the "Debris" name, it's classified as a Ship and has a Kerbal aboard — possibly mislabeled, or it's a stranded crewed stage.
- Stable LKO orbit (won't decay), but no mission listed.
- Nedory may need a recovery mission if there's no return capability.

Want me to check vessel details further or estimate a rescue Δv ?

You said: What are the current upgrade levels of my KSC buildings?

What are the current upgrade levels of my KSC buildings?

Claude responded: KSC building levels (0 = starting tier 1, max is typically 2 = tier 3):

Loaded tools, used ksp-mission-control integration

Loaded tools, used ksp-mission-control integration

KSC building levels (0 = starting tier 1, max is typically 2 = tier 3):

Building	Level	Tier
Mission Control	1	Tier 2
VAB	0	Tier 1
SPH	0	Tier 1
Launchpad	0	Tier 1

Runway	0	Tier 1
Tracking Station	0	Tier 1
R&D	0	Tier 1
Astronaut Complex	0	Tier 1
Administration	0	Tier 1

Constraints at current levels:

- VAB t1: 30-part limit, 18 t mass limit, no action groups, no subassemblies
- Launchpad t1: 18 t / 1.25 m height — caps your rocket size to small craft
- Tracking Station t1: No maneuver nodes, no patched conics — you have to eyeball intercepts
- R&D t1: Can't unlock nodes >100 sci, no surface samples allowed
- Astronaut Complex t1: Kerbals can't EVA in orbit, only in atmosphere/on ground; no flag-planting
- Mission Control t2: ✓ unlocked — you can accept more contracts now

Recommended next upgrades (Mun materials lab missions are your big science play):

1. Tracking Station → t2 to unlock maneuver nodes — essential for Mun transfer precision
2. R&D → t2 to lift the 100-sci cap (so you can bank Mun lab returns toward 160-cost nodes)
3. VAB + Launchpad → t2 to fly larger craft (e.g. Materials Lab payloads)

You said: Show me my full Kerbal roster.

Show me my full Kerbal roster.

Claude responded: 16 Kerbals total (15 available + 1 in flight).

Loaded tools, used ksp-mission-control integration

Loaded tools, used ksp-mission-control integration

16 Kerbals total (15 available + 1 in flight).

Core crew — Level 1 (all available)

Name	Specialty
Jebediah Kerman	Pilot

Valentina Kerman	Pilot
Bill Kerman	Engineer
Bob Kerman	Scientist

Trainee — Level 0

- Pawell Kerman — Pilot

Tourists — Level 0 (10, all on the ground waiting for rides)

Tamtrey, Albo, Rayfel, Dermal, Gersy, Ribler, Rorick, Sigsen, Kerbree, Thompely Kerman

In flight

- Nedory Kerman — aboard Nedory's Debris in 89×81 km Kerbin orbit (not in available roster)

Observations

- 10 tourists is a backlog — each one represents a contract worth funds/rep. Even a 1-tourist sub-orbital hop is good early income. With a 3-crew KV-3 pod you could clear them three at a time.
- Bob (Scientist) is critical for Mun science runs — scientists restore reset-able experiments (Goo, Materials Lab) for re-use across biomes, multiplying science return per mission.
- Nedory in LKO needs a plan: if the vessel has no return capability, you'll need a rescue. Could pair with a tourist contract (send tourists up, swap seats, bring Nedory home) for double value.
- Two veteran pilots (Jeb + Val) gives you SAS auto-pilot redundancy — useful since none of your current pods have built-in SAS.

You said: Give me the body info for Mun.

Give me the body info for Mun.

Claude responded: The Mun (Kerbin's moon)

Loaded tools, used ksp-mission-control integration

Loaded tools, used ksp-mission-control integration

The Mun (Kerbin's moon)

Property	Value
Mass	9.76×10^{20} kg
Radius	200 km

Atmosphere	None
Sphere of influence (SOI)	2,430 km
Semi-major axis from Kerbin	12,000 km
Orbital period	138,984 s (~6 h 27 m)
Rotation period	138,984 s — tidally locked

Derived stats

- Surface gravity: $\sim 1.63 \text{ m/s}^2$ ($\approx 0.166 \text{ g}$)
- Escape velocity (surface): $\sim 807 \text{ m/s}$
- Low Mun orbit velocity ($\sim 10 \text{ km}$): $\sim 570 \text{ m/s}$
- μ (GM): $\sim 6.51 \times 10^{10} \text{ m}^3/\text{s}^2$

Operational notes

- No atmosphere = no aerobraking and no parachutes; landings are pure propulsive. Budget $\sim 600\text{--}700 \text{ m/s}$ for a Mun landing burn from low orbit.
- Tidally locked — the same hemisphere always faces Kerbin. The Mun-facing side is ideal for direct CommNet to KSC; the far side needs a relay.
- Δv budget reference (LKO $\rightarrow$ return):
 - LKO $\rightarrow$ Mun transfer: $\sim 860 \text{ m/s}$
 - Mun capture: $\sim 310 \text{ m/s}$
 - Land from low Mun orbit: $\sim 580 \text{ m/s}$
 - Ascent + return to Kerbin SOI: $\sim 860 \text{ m/s}$
 - Total round trip: $\sim 2,600 \text{ m/s}$ + reentry margin

Want me to compute a specific transfer (e.g. LKO $\rightarrow$ low Mun Hohmann) or check synchronous Mun orbit altitude?

You said: Body info for Kerbin.

Body info for Kerbin.

Claude responded: Kerbin

get_body_info

get_body_info

Kerbin

Property

Value

Mass	5.29×10^{22} kg
Radius	600 km
Atmosphere height	70 km
Sphere of influence	~84.16 million km
Semi-major axis (from Sun)	13.6 Gm (~1 AU equivalent)
Orbital period	9,203,544 s (~106.5 days = 1 Kerbin year)
Rotation period	21,549 s (~5 h 59 m 9 s = 1 Kerbin day)

Derived stats

- μ (GM): 3.531×10^{12} m³/s²
- Surface gravity: 9.81 m/s² (essentially 1 g)
- Escape velocity (surface): ~3,431 m/s
- Low Kerbin orbit velocity (~80 km): ~2,279 m/s
- Keostationary orbit altitude: ~2,863 km (verified earlier)

Operational notes

- Atmosphere extends to 70 km — orbit above this for stable flight; below ~70 km you'll experience drag and eventual reentry. Aerobraking and parachutes are viable here.
- One day $\approx$ 6 hours — keostationary altitude is much lower than Earth's GEO (~36,000 km) because of the fast rotation.
- SOI of 84 million km is comfortably outside the Mun's orbit (12 Mm) and Minmus's (47 Mm), so both are deep inside Kerbin's gravity well.
- Δv to reach orbit: ~3,400 m/s from the launchpad (gravity + drag losses included), making lsp 300 / mass-ratio 5 rockets (yielding ~4,735 m/s as we calculated) a comfortable single-stage-to-orbit budget — though staging is usually still more efficient.

You said: What are the difficulty settings for my current career?

What are the difficulty settings for my current career?

Claude responded: Career difficulty: Normal (all multipliers at 1×).

Loaded tools, used ksp-mission-control integration

Loaded tools, used ksp-mission-control integration

Career difficulty: Normal (all multipliers at 1×).

CommNet

- Enabled, range modifier 1×, DSN modifier 1×
- Bodies occlude signal (line-of-sight required)
- Signal NOT required for control — you can still pilot probes without comms, just won't get science transmission
- No plasma blackout during reentry

Rewards / penalties (all 1×)

- Science, Funds, Reputation rewards: 1×
- Funds, Reputation penalties: 1×

Physics / gameplay

- Reentry heating: 1× (default)
- Missing crews respawn: enabled — lost Kerbals return to the roster after a delay

Net assessment

This is the standard balanced career setting — moderate challenge, no science/funds boost or handicap. The kindest setting in play is missing crews respawn: losing a Kerbal isn't permanent (so Nedory in orbit isn't a hard-loss risk if things go wrong, but you'd lose contracts and rep).

The strictest setting in play is occludeBodies: behind a moon or planet you lose signal, so the Mun far side needs a relay if you want science transmission from there. Not a problem at your current science-tier though — manual control still works without signal.

You said: I want to get a satellite into a synchronous orbit around Kerbin.

I want to get a satellite into a synchronous orbit around Kerbin. Based on my current tech tree and available parts, what engines and fuel tanks should I consider, and roughly how much delta-v will I need for a launch to orbit plus the transfer?

Claude responded: Let me compute the exact transfer Δv first.

Let me compute the exact transfer Δv first.

calculate_hohmann_transfer

calculate_hohmann_transfer

Δv budget

Phase	Δv
Launchpad → LKO (~80 km)	~3,400 m/s
Hohmann burn 1 (LKO → transfer ellipse)	668 m/s
Hohmann burn 2 (circularize at keo)	431 m/s

Inclination cleanup + margin

~200 m/s

Total

~4,700 m/s

Transfer coast time: ~5,000 s (~83 min).

Engine recommendations from your unlocked parts

Booster stage (atmospheric ascent) — want high thrust, decent sea-level Isp:

- LV-T45 "Swivel" (215 kN, 320/250 s, gimbaled) — your steering core. The gimbal matters because none of your pods have built-in SAS.
- Pair with 2× RT-10 "Hammers" or 2× BACC "Thumpers" as radial SRBs for TWR off the pad. Hammers are cheap and respect the 18 t launchpad limit.
- If you want a 2.5 m core instead: RE-I5 "Skipper" (650 kN, 320/280 s) — but that pushes you near the 18 t launchpad cap quickly.

Upper stage / transfer stage (vacuum) — want high Isp, low mass:

- LV-909 "Terrier" (60 kN, 345 s vac, only 0.5 t) — ideal for a light comsat. 60 kN is plenty to push a ~2 t upper stage.
- RE-L10 "Poodle" (250 kN, 350 s vac) — best Isp you have, but at 1.75 t it's overkill for a tiny satellite.

Fuel tank recommendations

- Upper/transfer stage: FL-T200 (1.13 t wet, 200 units) with a Terrier gives ~2,000 m/s Δv on a ~0.5 t payload — comfortable margin for the 1,099 m/s transfer.
- Booster stage: FL-T800 (4.5 t wet) under a Swivel works for the lift portion, or use X200-8 on a 2.5 m core if you go Skipper.

Sample two-stage design (rough mass budget, well under 18 t):

Payload : Stayputnik + battery + HG-5 antenna + advSAS module ~0.4 t

Transfer : LV-909 Terrier + FL-T200 ~1.7 t

Booster : LV-T45 Swivel + FL-T800 + 2× Hammer SRBs ~12 t

Total ~14 t

Practical considerations from your career state

- Antenna: Use the HG-5 (5 Mm range) from Basic Science — combined with DSN level 1 (Tracking Station t1, 2 Gm), gives you ~100,000 km combined range. Plenty for a 3,463 km keo sat. The internal 5 km probe antenna won't work at that altitude.
- No solar panels yet — Electrics (90 sci) is locked. A pure battery-powered comsat will drain over time. Two workarounds:
 1. Add several battery packs (cheap fix, finite lifetime)
 2. Send a Kerbal pilot in a Mk1 pod instead of Stayputnik (50 EC + crew SAS, no ongoing draw beyond probe core baseline)
- No maneuver nodes (Tracking Station t1) — you'll have to time burns by eye against orbital periods. Keostationary is particularly unforgiving on circularization timing.
- Launchpad 18 t limit — keeps you in 1.25 m stack territory; the design above fits comfortably.

Recommendation

Before launching, consider upgrading Tracking Station to t2 (38k funds) — maneuver nodes will save you 100+ m/s in sloppy circularization. You have ~504k funds, so the upgrade is affordable. Without it, plan an extra ~200 m/s of margin (already included above).